Does Feed Curation Shape What Readers Know?

A Controlled Experiment on Algorithmic News Exposure

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***Abstract*—We investigate how algorithmic feed curation affects belief formation following a crisis event with evolving facts. Using the assassination attempt on Slovak Prime Minister Robert Fico (May 15, 2024) as a case study, we scored 64 English-language news articles across six axes—evidence and attribution, uncertainty signaling, context completeness, affective intensity, conflict framing, and call to action—and constructed three counterfactual feeds optimizing for recency, engagement, and epistemic integrity. Each feed was deployed as a realistic X–style interface. Survey participants were randomly assigned to one feed condition and subsequently assessed on factual accuracy, confidence calibration, and subjective impressions of the feed. We find that participants who were assigned the engagement-optimized feeds are more confidently wrong in their answers about factual true/false questions. In addition, the participants across all three feeds do not perceive the feeds to be different in meaningful ways, like trustworthiness, perceived bias, emotional activation, perceived contradictions, charged language, or contextual depth. Our results suggest that the ranking objective of a news feed affects how confident a user is about false beliefs following an event, in a way that happens imperceptibly to the user.**

***Index Terms*—algorithmic curation, belief formation, news feeds, epistemic integrity, media effects, crisis communication**

I. Introduction

When a crisis breaks—a shooting, a natural disaster, a contested political event—the facts are rarely settled within the first hours. Early reports are fragmentary, sourcing is uncertain, and responsible journalists hedge their claims. Yet this is precisely the window in which the public forms its initial beliefs, beliefs that are subsequently resistant to correction [1]. The medium through which people encounter these early reports is increasingly an algorithmically curated feed: a ranked, filtered, personalized selection of content optimized for some objective function, whether engagement, recency, or editorial judgment.

A growing body of research demonstrates that algorithmic curation affects exposure [3], polarization [4], and emotional contagion [5]. However, less work has examined a more fundamental question: does the *ranking objective* of a feed—what it optimizes for—change what users *factually believe* about a real-world event? This is distinct from the question of whether feeds affect attitudes or opinions. We are interested in whether curation shapes the accuracy and calibration of beliefs about verifiable facts.

We were motivated by several recent cases in which the same underlying event produced dramatically different public narratives depending on the platform and curation logic. The media coverage of U.S. Border Patrol agents shooting and killing Alex Pretti on January 24, 2026, for instance, generated simultaneously circulating content that included official condemnation framing the subject incorrectly as a domestic terrorist, video compilations revealing contradictory official statements, and AI-manipulated imagery that was shared alongside legitimate reporting without distinction. Similar dynamics emerged in the aftermath of U.S. Immigration and Customs Enforcement officers shooting and killing Renée Good on January 7, 2026. The information environment fractured along platform and curation lines. In each case, the *same underlying facts* produced different beliefs in different audiences, not just because of fabricated content, but because of what was surfaced, in what order, with what framing.

This paper presents an experimental study of this phenomenon. We select a real crisis event, the assassination attempt on Slovak Prime Minister Robert Fico on May 15, 2024, and construct three counterfactual news feeds from authentic English-language coverage, with each feed optimizing for a different objective. We then measure participants' factual beliefs, confidence, and subjective impressions after exposure to one of these feeds. Our contributions are: (1) a reproducible codebook and scoring pipeline for multi-axis article assessment, (2) three formally defined feed-ranking algorithms grounded in the communication and information-quality literature, (3) a realistic feed-delivery platform with integrated survey instrumentation, and (4) empirical measurements of how feed curation affects belief accuracy and calibration and a framework with which to analyze them.

II. Related Work

The impact of design choices in any one algorithm on the digital information ecosystem is well documented, especially how ranking influences user perception. This phenomenon connects to broader work on algorithmic exposure diversity [3] and political polarization [4], with Pariser [16] noting that personalization can isolate users from disconfirming information. Such bubbles can exacerbate political environments. Notably, Bail et al. (2018) [4] found that exposure to opposing views on social media can actually increase, and not decrease, political entrenchment. Further, social platforms has also proven to facilitate emotional contagion. Kramer et al. (2014) [5] ran an experiment in which the emotional valence of a user's feed significantly shifted that user's mood and posting behavior.

Misinformation is the most critical barrier to overcome when we face factual crises in this age of online information. Lewandowsky et al. (2012) [1] highlight the ”continued influence effect”: where false information will continue to shape a person’s perception even after a correction is provided. This effect points to the importance of the initial ”window of exposure” provided by any feed providing news on current events. To measure a user’s resulting belief, we turn to the literature on calibrations in judgment. Brier (1950) [9] introduced the Brier score as a metric for the accuracy of probabilistic predictions. Tetlock (2005) [10] later expands upon this in his study of expert political judgment. For our experiment, we utilize Brier scores to assess how well a participant’s confidence aligns with the the objective truth. In short, we are trying to understand the epistemic costs of different algorithm curation objectives.

III. METHODS

A. Event Selection

We required a candidate event satisfying three criteria. First, the event must have *low salience in the American consciousness* to minimize prior beliefs among our predominantly U.S.-based survey participants. Second, it must have *high English-language media coverage* to provide a sufficiently large and diverse corpus. We specifically want to avoid any nuances that could be lost in translation. Third, there must be *genuine factual uncertainty within the first 24–48 hours*—evolving reports, conflicting accounts, and information that was unknown at the time of early coverage but later confirmed or denied.

We evaluated four candidate events across different domains: the Hungary deepfake ads in the 2026 election (politics), the Alabama IVF/frozen embryo ruling of February 2024 (health), the Kanan Church airstrike of January 2024 (violence), and the Coupang data breach of November 2025 (safety). We selected the assassination attempt on Slovak Prime Minister Robert Fico, which occurred on May 15, 2024 in Handlová, Slovakia. Fico was shot multiple times after a government cabinet meeting. The event generated extensive global English-language coverage, was not widely followed in the United States, and featured genuine early uncertainty regarding the number of shots fired, Fico’s medical condition, and the suspect’s identity and motive.

B. Corpus Construction

We queried the GDELT 2.0 (Global Database of Events, Language, and Tone) for English-language articles published within the first 24 hours following the shooting. This yielded an initial set of several hundred articles. We applied the following filters:

1) **Paywall removal:** Articles behind hard paywalls were excluded, since survey participants must be able to access the full text when clicking through from the feed.
2) **Deduplication:** Many outlets republish identical wire-service copy (AP, Reuters). We identified and removed duplicates by comparing article text, retaining only one version per substantively distinct piece of coverage.
3) **Link verification:** Each article URL was manually verified to confirm it resolved to a live, accessible page. Articles with dead links were removed from the corpus.

The final corpus consists of 64 articles from outlets spanning 15 countries, including the United States, United Kingdom, India, Turkey, Qatar, Pakistan, Australia, New Zealand, Singapore, Indonesia, Nepal, Bangladesh, Armenia, Lithuania, Ghana, South Africa, and Germany. This geographic diversity provides natural variation in framing, sourcing, and editorial tone.

C. Article Scoring

Each article was scored on six axes adapted from established frameworks in journalism and communications research. The following axes were scored, on a 1-3 scale.

- **Evidence & Attribution (EA).** Is the article grounded in verifiable evidence with clear source attribution? [11]
- **Uncertainty Signaling (US).** Does the article communicate that facts are evolving? [12]
- **Context Completeness (CC).** How much background and timeline does the article provide? [13]
- **Affective Intensity (AI).** Does the article use emotional language that drives engagement? [14]
- **Conflict Framing (CF).** Does the article frame the event in terms of opposing groups or assign blame? [13]
- **Call to Action (CA).** Does the article push behavior rather than understanding? [15]

The complete codebook can be found in the project’s GitHub repo.

1) Codebook Development and Validation: The codebook was developed through an iterative process designed to decompose complex editorial concepts into objective, binary observations. For each axis, we identified two binary factors; an article receives a score of 1 if neither factor is present, 2 if exactly one factor is present, and 3 if both factors are present.

To validate the codebook, we employed an independent double-coding procedure. Two team members independently scored a selected subset of ten articles across all axes, without any prior discussion of individual ratings. Inter-rater reliability was assessed using weighted Cohen’s κ [19], with quadratic weights applied to account for the ordinal nature of the 1–3 scale—penalizing larger discrepancies more heavily than smaller ones:

$$w_{ij} = 1 - \frac{(i-j)^2}{(k-1)^2}, \quad \kappa_w = \frac{P_o - P_e}{1 - P_e} \qquad (1)$$

where $k = 3$, $P_o = \sum_{ij} w_{ij}\, o_{ij}$ is the weighted observed agreement, and $P_e = \sum_{ij} w_{ij}\, e_{ij}$ is the weighted expected agreement under independence.

We set a minimum reliability threshold of $\kappa_w > 0.70$, consistent with the “substantial agreement” benchmark established by [18]. Table I reports κ_w for each axis.

During the first iteration, the overall pooled κ_w fell below 0.70, particularly for AI ($\kappa_w = 0.60$) and CF ($\kappa_w = 0.64$). Disagreements were resolved through structured adjudication.

TABLE I
WEIGHTED COHEN'S KAPPA (κ_w) BY CODING AXIS

Axis	Weighted κ_w	Interpretation
Evidence & Attribution (EA)	0.750	Substantial
Uncertainty Signaling (US)	0.932	Almost perfect
Context Completeness (CC)	1.000	Perfect
Affective Intensity (AI)	0.600	Moderate
Conflict Framing (CF)	0.643	Substantial
Call to Action (CA)	—	N/A (no variance)
Overall (pooled)	**0.861**	**Almost perfect**

Note. Almost perfect / Perfect ($\kappa_w \geq 0.80$). Substantial ($0.70 \leq \kappa_w < 0.80$). Below threshold ($\kappa_w < 0.70$). Quadratic weighting scheme; $k = 3$ rating levels. Interpretation follows [18]. "Call to Action" received identical ratings of 1 from both coders across all articles, precluding variance-based computation.

For instance, disagreements were traced to ambiguities in the boundary between rating levels 1 and 2. Following refinements on the codebook and a secondary iteration on a new held-out sample, all axes pooled met the reliability threshold, achieving a final pooled $\kappa_w = 0.86$. The CA axis yielded identical ratings of 1 across all ten articles, precluding variance-based computation, representing perfect consensus agreement.

2) LLM-Assisted Scoring: To scale the annotation process across all 64 articles, we developed an automated scoring pipeline with Claude 3.5 Sonnet. This model was selected for its high performance in complex reasoning tasks, so that it could replicate replicates human scoring logic established in the codebook. [20]

The LLM was provided with a context rich prompt consisting of two components:

1) The full codebook: Detailed definitions of all six axes, including the specific binary factors for each score.
2) Adjudicated examples: 10 human-labeled examples for few-shot dependency.

After the receiving the scores for all articles, two team members cross-checked some articles randomly to confirm that the scores seem reasonable. Unfortunately, we do not have the data to confirm a weighted Cohen's kappa between the human annotators and the LLM.

D. Feed Construction

Using the six-axis scores, we constructed three counterfactual feeds, each implementing a different ranking objective.

1) Recency Feed: The recency feed ranks articles in reverse chronological order by publication timestamp. The feed displays the 12 most recent articles 24 hours after the shooting, ordered newest-first. This feed is deterministic and does not change between participants.

2) Engagement Feed: The engagement feed ranks articles by predicted engagement. We collected real engagement metrics (total interactions across Facebook, X/Twitter, Reddit, and Pinterest) using the Buzzsumo Content Analyzer for each article in the corpus. Articles were ranked in descending order of total engagement, and the top 12 were selected.

3) Epistemic Integrity Feed: The epistemic integrity feed represents our normative condition: a feed optimized for an informed citizenry. Each article receives an epistemic integrity score:

$$\text{epistemic} = \alpha\cdot\text{EA} + \beta\cdot\text{US} + \gamma\cdot\text{CC} - \delta\cdot\text{AI} - \eta\cdot\text{CF} - \xi\cdot\text{CA} \quad (2)$$

with $\alpha = 1.5$, $\beta = 1.0$, $\gamma = 1.2$, $\delta = 1.0$, $\eta = 1.0$, $\xi = 0.8$. The weights reflect our judgment that primary evidence matters most (1.5), followed by contextual completeness (1.2) and uncertainty signaling (1.0), while affective intensity (-1.0) conflict framing (-1.0), and mobilization (-0.8) are penalized. Unlike the other two feeds, the epistemic feed uses *weighted random sampling*: on each page load, 12 articles are drawn without replacement with probability proportional to the squared epistemic score (shifted to ensure positivity). This introduces controlled variation across participants while consistently favoring high-quality coverage.

4) Advertisements: To increase ecological validity, each feed includes two "promoted" posts styled as Twitter advertisements. These were selected to be contemporaneous with the May 2024 coverage period: Max/Netflix, Paris 2024 Olympics, Taylor Swift's Eras Tour European dates, Apple iPad Pro (M4 launch), UEFA Euro 2024. Ads are inserted at positions 4 and 9 in the feed to mimic organic ad placement on social platforms.

E. Feed Deployment

Each feed was deployed as a standalone HTML page on Netlify, replicating the visual language of Twitter/X with outlet avatars, timestamps, and engagement metrics. Clicking a post opens the original article in a new tab to read the full text. This design decision—linking to actual articles rather than displaying excerpts—was made to simulate how news is actually consumed on social media: headlines and snippets in the feed, with click-through for full reading.

The three feeds are accessible at separate Netlify URLs, enabling random assignment by distributing different links to different participants. All rendering, scoring, and sampling logic runs client-side in the browser; no server-side code or database is required. The complete source code, article data (including all 64 scored articles with URLs), and deployment configuration are available in our public repository.[1]

F. Survey Design

The survey is integrated into each feed page as a multi-page instrument that wraps around the feed. Participants progress through eight sequential pages. The HTML files for the surveys, including the 10 true/false questions and answers, can be found in the GitHub repo.

Page 1: Screening (Q1–Q4). Participants report their prior familiarity with the Fico shooting (5-point scale), whether they had followed coverage previously (Yes / No / Vague memory), their primary news source (social media, TV, online news, newspapers, podcasts, or none), and how often they follow international news (Never–Always).

[1] GitHub repo: https://github.com/inthree3/epistemic-feed-study

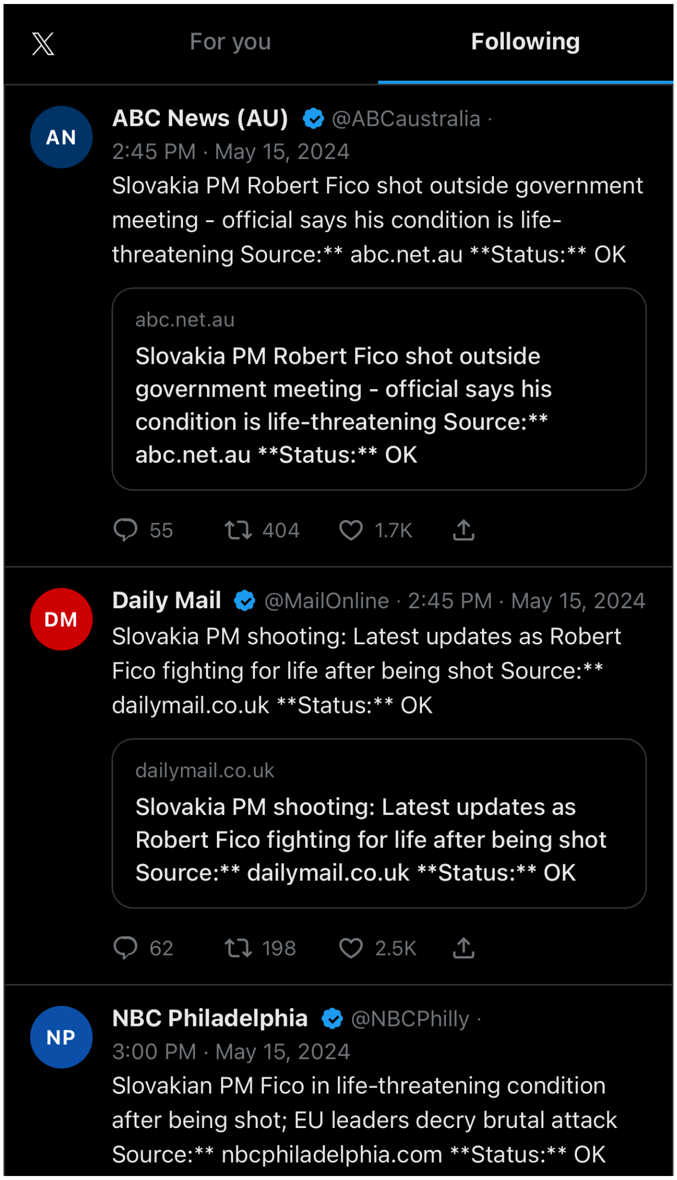

Fig. 1. Sample screenshot of recency feed

Page 2: Feed Exposure (Q5–Q6). The Twitter-style feed is displayed in a scrollable container. After scrolling, participants report how many posts they read in full (0, 1–3, 4–6, 7–10) and whether any posts felt like advertisements (Yes / No / Unsure).

Page 3: Belief Accuracy and Confidence (Q7–Q16). Participants respond to 10 factual statements about the event, selecting True, False, or Not Enough Information (NEI) for each. Each response is accompanied by a confidence slider (0–100). The 10 statements were constructed to span key factual dimensions of the event: location, suspect apprehension, number of shots, political context, official characterization of motive, Fico's survival, suspect demographics, weapon legality, international reactions, and Fico's tenure history. Some questions, including number of shots, were designed to capture aspects of the situation that may not have been immediately obvious following the incident.

Page 4-6: Subjective Impressions/Content Assessment (Q17–Q23). Six 8-point Likert scales (0–7) measuring perceived trustworthiness, perceived bias, emotional activation, perceived contradiction between posts, emotional charge of language, and amount of background context provided. Included an attention check question to identify inattentive respondents.

Page 7: Demographics (Q24–Q27). Age bracket, education level, country of residence, and political orientation (7-point liberal–conservative scale with a "prefer not to say" option).

Page 8: Debriefing. Participants are informed that the study examined how feed curation affects understanding, that the event was real, and that Fico survived.

1) Response Collection: Survey responses are collected via Netlify forms and exported via CSV for analysis. We make our analysis available in the GitHub repository linked in the Appendix.

G. Participant Recruitment

Participants were solicited from the authors' social and academic circles. Due to timing constraints, no participants were solicited through Prolific. All participants are 16 years or older, and all are fluent in English.

H. Outcome Measures

For each participant i, we define three primary outcome variables.

Belief Accuracy. The proportion of correctly answered factual questions:

$$A_i = \frac{1}{10} \sum_{j=1}^{10} \text{correctness}_{ij} \tag{3}$$

where $\text{correctness}_{ij} \in \{0, 1\}$ based on agreement with the ground-truth answer (True, False, or Not Enough Information).

Overconfidence (Brier Score). A calibration measure capturing the squared difference between expressed confidence and actual correctness:

$$\text{Brier}_i = \frac{1}{10} \sum_{j=1}^{10} (p_{ij} - y_{ij})^2 \tag{4}$$

where $p_{ij} = \text{confidence}_{ij}/100 \in [0, 1]$ and $y_{ij} = \text{correctness}_{ij} \in \{0, 1\}$. When a participant selects "Not Enough Information," we assign $p_{ij} = 0.5$, reflecting maximal uncertainty; this yields a neutral Brier contribution of $(0.5 - y_{ij})^2 \leq 0.25$, which neither heavily penalizes nor rewards epistemic abstention.A lower Brier score indicates better calibration. Notably, a "Not Enough Information" results in a neutral score of 0.25. We hypothesize that the engagement feed produces higher Brier scores (overconfidence: high confidence paired with lower accuracy) relative to the epistemic feed.

Selection rate of "Not Enough Information." The proportion of questions for which participant i selected "Not Enough Information" rather than True or False:

$$\text{NEI}_i = \frac{1}{10}\sum_{j=1}^{10} \mathbb{1}[\text{response}_{ij} = \text{NEI}] \quad (5)$$

where $\mathbb{1}[\cdot]$ is the indicator function, equal to 1 when the condition holds and 0 otherwise. A higher NEI rate reflects greater epistemic humility — a willingness to acknowledge the limits of one's knowledge rather than guess. We hypothesize that the engagement feed suppresses NEI selection relative to the epistemic feed, as definitively framed articles may create a false sense of certainty even when the underlying facts are ambiguous.

I. Statistical Analysis

All statistical analyses are conducted in Python using `scipy` and `statsmodels`. For each primary outcome—belief accuracy, Brier score, and NEI selection rate—we first run a one-way ANOVA to test whether any feed condition differs significantly from the others. Where the ANOVA reveals a significant or near-significant effect, we follow up with pairwise Welch's independent-samples t-tests, which do not assume equal variances across groups and are appropriate given the unequal cell sizes in our design ($n_{\text{Epistemic}} = 23$, $n_{\text{Recency}} = 19$, $n_{\text{Engagement}} = 19$). All t-tests are two-sided with $\alpha = 0.05$.

Effect sizes are reported as Cohen's d, computed as:

$$d = \frac{M_1 - M_2}{SD_{\text{pooled}}}, \quad SD_{\text{pooled}} = \sqrt{\frac{SD_1^2 + SD_2^2}{2}} \quad (6)$$

We interpret d using conventional benchmarks: small ≥ 0.2, medium ≥ 0.5, large ≥ 0.8 [19]. Degrees of freedom for Welch's t-tests are estimated using the Welch–Satterthwaite equation and are reported as non-integer values.

Secondary outcomes (perceived trustworthiness, bias, emotional activation, contradictions, charged language, and contextual depth; Q17–Q22) are analyzed using one-way ANOVAs. Perception items are rated on a 0–7 scale.

Participants who failed the instructed-response attention check (Q23) are excluded from all analyses prior to any statistical testing ($n = 9$ excluded, 12.5% of total responses). An additional 2 participants were excluded for failing to provide answers to any factual verification question, yielding a final analytic sample of $n = 61$. No participants reported prior familiarity with the Fico shooting (mean familiarity = 1.12 on a 1–5 scale; 0% rated 4 or 5), so no exclusions on the basis of prior knowledge are necessary.

We conduct two pre-specified robustness checks. First, we exclude participants who reported reading zero posts ($n = 11$, 18.0% of the analytic sample), leaving $n = 50$ readers, to test whether findings are driven by non-engagement rather than feed curation. Second, we test a feed type $\times$ posts-read interaction using ordinary least squares regression to assess whether feed effects shrink as reading depth increases, consistent with a salience-over-availability account of curation effects.

IV. Results

A. Participants and Exclusions

Of 72 total responses collected, 9 (12.5%) failed the instructed-response attention check (Q23) and 2 did not select any answers. All were excluded prior to any analysis, yielding an analytic sample of $n = 61$. Feed conditions were roughly balanced after exclusion: Epistemic ($n = 23$), Engagement ($n = 19$), and Recency ($n = 19$).

No participants reported prior familiarity with the Fico shooting. Mean familiarity was 1.12 ($SD = 0.40$) on a 1–5 scale (1 = not at all familiar), and 0% of participants rated their familiarity at 4 or 5. No participant reported having followed prior coverage of the event. This confirms the effectiveness of our event selection criteria: participants entered the study without pre-existing beliefs about the target event that could confound feed condition effects on belief formation.

The analytic sample was predominantly young and highly educated: 42.6% were aged 18–24 and 39.3% aged 25–34, with 82.0% holding at least a bachelor's degree. The majority were US-based (82.0%), with 18.0% international (South Korea, India, Norway). Primary news sources were social media (52.5%) and online news sites (34.4%). The sample skewed politically liberal (mean 2.98 on a 1–7 scale where 4 = moderate). These characteristics should be considered when assessing generalizability: findings may be most applicable to young, digitally-engaged, educated news consumers.

B. Primary Outcomes

1) Belief Accuracy: Table II reports descriptive statistics for all primary outcomes by feed condition. Mean belief accuracy was 0.470 ($SD = 0.267$) for the Epistemic condition, 0.453 ($SD = 0.171$) for Engagement, and 0.340 ($SD = 0.186$) for Recency. A one-way ANOVA revealed a non-significant trend toward a feed condition effect on accuracy, $F(2, 58) = 2.12$, $p = .130$.

Pairwise Welch's t-tests did not reveal any significant comparisons. Engagement participants scored higher on accuracy than Recency participants, though this difference did not reach significance, $t(35.7) = 1.94$, $p = .060$, $d = 0.63$, a medium-to-large trend. The Epistemic versus Recency comparison similarly approached but did not reach significance, $t(39.0) = 1.85$, $p = .072$, $d = 0.56$. Epistemic and Engagement conditions did not differ on accuracy, $t(37.9) = 0.25$, $p = .805$, $d = 0.08$.

2) Belief Calibration (Brier Score): The Brier score is our primary calibration measure, capturing the alignment between expressed confidence and actual accuracy. Lower scores indicate better calibration. Mean Brier scores were 0.212 ($SD = 0.060$) for Epistemic, 0.274 ($SD = 0.073$) for Engagement, and 0.234 ($SD = 0.060$) for Recency. A one-way ANOVA revealed a significant effect of feed condition on calibration, $F(2, 58) = 4.88$, $p = .011$.

TABLE II
DESCRIPTIVE STATISTICS FOR PRIMARY OUTCOMES BY FEED CONDITION ($n = 61$). VALUES ARE MEAN (SD).

Outcome	Epistemic ($n = 23$)	Recency ($n = 19$)	Engagement ($n = 19$)
Belief Accuracy ↑	0.470 (0.267)	0.340 (0.186)	0.453 (0.171)
Brier Score ↓	0.212 (0.060)	0.234 (0.060)	0.274 (0.073)
NEI Selection Rate	0.365 (0.287)	0.521 (0.280)	0.337 (0.263)

↑ higher is better; ↓ lower is better

TABLE III
DESCRIPTIVE STATISTICS FOR SECONDARY OUTCOMES (PERCEIVED FEED CHARACTERISTICS) BY FEED CONDITION ($n = 61$). ALL ITEMS RATED 0–7. NO SIGNIFICANT DIFFERENCES WERE OBSERVED ACROSS CONDITIONS (ALL $p > .10$).

Measure	Epistemic	Recency	Engagement	p
Trustworthiness	4.17 (1.27)	4.16 (1.74)	5.00 (1.56)	.149
Perceived Bias	3.35 (1.47)	2.95 (1.87)	2.84 (1.54)	.564
Emotional Activation	1.91 (1.41)	1.32 (1.73)	2.05 (1.90)	.356
Contradictions	2.74 (1.51)	2.11 (1.41)	2.63 (2.09)	.450
Charged Language	3.61 (1.95)	2.79 (1.75)	2.47 (1.78)	.123
Contextual Depth	3.65 (1.67)	2.63 (1.42)	2.89 (1.66)	.104

Pairwise comparisons identified the Epistemic versus Engagement contrast as the primary driver of this effect. Epistemic participants were significantly better calibrated than Engagement participants, $t(34.8) = -2.96$, $p = .005$, $d = 0.93$,a large effect. The Epistemic versus Recency comparison did not reach significance, $t(38.5) = -1.16$, $p = .255$, $d = 0.36$, nor did the Engagement versus Recency comparison, $t(34.7) = 1.86$, $p = .071$, $d = 0.60$.

Figure 2 displays mean accuracy, Brier score, and NEI selection rate by feed condition with standard error bars and significance annotations.

3) NEI Selection Rate: The NEI selection rate reflects the proportion of questions for which participants expressed maximal uncertainty rather than committing to a true or false response. Mean NEI rates were 0.365 ($SD = 0.287$) for Epistemic, 0.337 ($SD = 0.263$) for Engagement, and 0.521 ($SD = 0.280$) for Recency. A one-way ANOVA revealed a near-significant effect of feed condition, $F(2, 58) = 2.47$, $p = .093$.

Pairwise comparisons revealed that Engagement participants selected NEI significantly less often than Recency participants, $t(35.9) = -2.09$, $p = .044$, $d = 0.68$, a large effect. The Epistemic versus Recency comparison approached but did not reach significance, $t(38.9) = -1.78$, $p = .084$, $d = 0.55$. Epistemic and Engagement conditions did not differ on NEI rate, $t(39.5) = 0.33$, $p = .740$, $d = 0.10$.

C. Secondary Outcomes

Table III reports means and standard deviations for all six secondary perception measures. None reached statistical significance in one-way ANOVAs, indicating that participants did not perceive meaningful differences between feed conditions on any perceptual dimension.

The absence of significant perceptual differences is notable given the substantial calibration differences observed in the primary outcomes. Participants in the Engagement condition did not perceive their feed as more emotionally charged, more biased, or less trustworthy than participants in the Epistemic condition, despite being significantly worse calibrated. This suggests that feed curation effects on belief formation operate below the threshold of conscious awareness, limiting the effectiveness of individual media literacy as a corrective mechanism.

Two perception measures showed directional patterns approaching but not reaching significance. Epistemic participants reported the highest contextual depth ($M = 3.65$) compared to Recency ($M = 2.63$) and Engagement ($M = 2.89$), $F(2, 58) = 2.36$, $p = .104$, consistent with our feed design. Engagement participants rated their feed as most trustworthy ($M = 5.00$) despite producing the worst calibration, $F(2, 58) = 1.97$, $p = .149$, a pattern consistent with the overconfidence account.

D. Robustness Checks

1) Sensitivity Analysis: Readers Only: 10 participants (16.4%) reported reading zero posts. The Recency condition had a disproportionately high non-reader rate (31.6%) relative to Epistemic (8.7%) and Engagement (10.5%), though this difference was not statistically significant, $\chi^2(6) = 7.57$, $p = .272$. To assess whether Recency's lower accuracy reflects non-engagement rather than a genuine curation effect, we replicated the primary analyses excluding non-readers ($n = 50$: Epistemic $n = 21$, Engagement $n = 16$, Recency $n = 13$).

The primary Brier score finding was robust to this exclusion. Among readers, the Epistemic versus Engagement calibration advantage remained significant, $t(28.0) = -2.37$, $p = .025$, $d = 0.80$. A new significant finding also emerged among readers: the Engagement versus Recency Brier comparison reached significance, $t(26.8) = 2.10$, $p = .045$, $d = 0.77$, indicating that among participants who actually read posts, the engagement feed produced worse calibration than the recency feed as well. Notably, Recency accuracy remained low among actual readers ($M = 0.350$, $SD = 0.171$), close to the full-sample mean ($M = 0.340$), indicating that the Recency condition's poor accuracy reflects a genuine curation effect rather than an artifact of non-engagement. The Engagement versus Recency accuracy advantage strengthened among readers, $t(21.3) = 2.53$, $p = .019$, $d = 0.96$.

2) Reading Depth Interaction: We tested whether feed effects on calibration shrink as reading depth increases, using an OLS regression with feed condition, posts-read (ordinal: 0–3), and their interaction as predictors. The Epistemic × posts-read interaction for Brier score approached but did not reach significance, $b = 0.048$, $p = .067$, $R^2 = .189$. The direction of the coefficient is consistent with a salience-over-availability account: feed effects are largest among light readers who encounter only the top-ranked articles, and diminish as reading depth increases and participants approach the same underlying

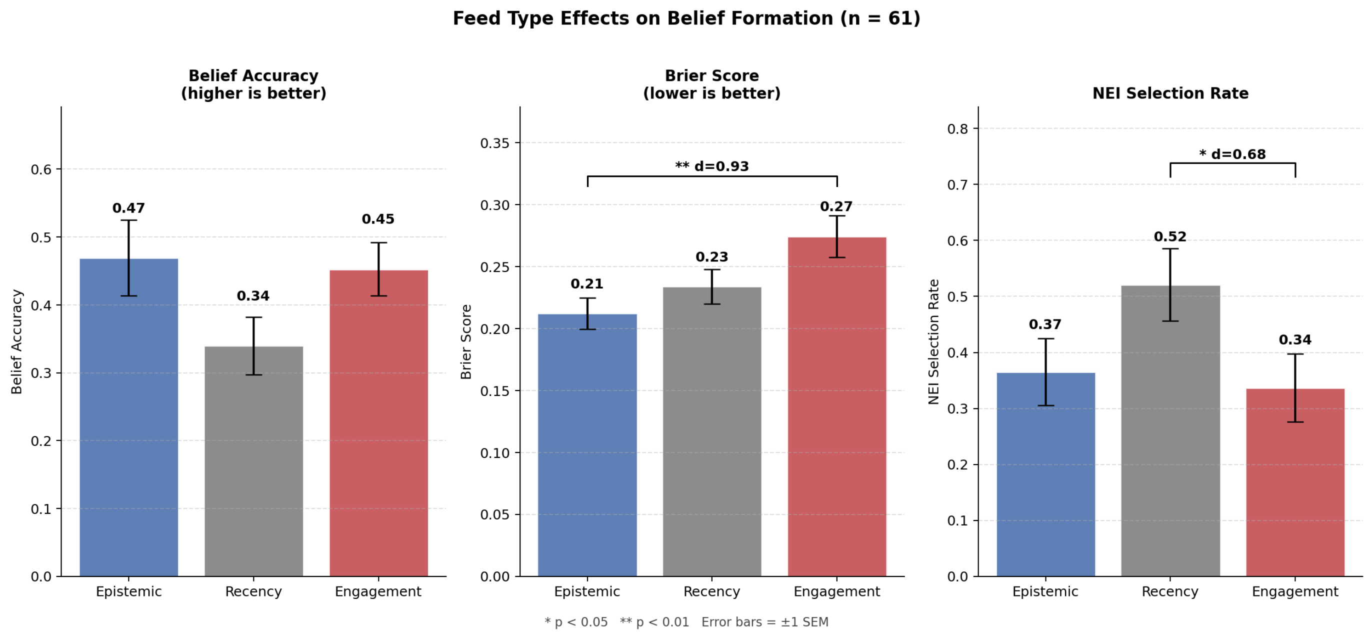

Fig. 2. Mean belief accuracy, Brier score, and NEI selection rate by feed condition ($n = 61$). Error bars represent ± 1 SEM. Significance brackets indicate Welch's t-test results: * $p < .05$, ** $p < .01$. Brier score: lower is better. NEI selection rate: proportion of questions answered with "Not Enough Information."

information regardless of ranking. This remains an exploratory finding requiring larger samples and richer engagement instrumentation to confirm.

V. Discussion

The results provide partial support for our central hypothesis that feed curation objectives shape factual belief formation. Feed condition did not produce a significant overall effect on raw belief accuracy, $F(2, 58) = 2.12$, $p = .130$, though a directional pattern emerged: the Recency condition produced the lowest mean accuracy ($M = 0.340$), with the Epistemic condition highest ($M = 0.470$) and Engagement intermediate ($M = 0.453$). No pairwise accuracy comparisons reached conventional significance in the full sample, though the Engagement versus Recency difference showed a medium-to-large trend ($p = .060$, $d = 0.63$) that strengthened among readers who reported engaging with at least one post ($d = 0.96$, $p = .019$). The Recency feed's poor performance is consistent with the structure of breaking news coverage, as articles published in a 24-hour window tend to be the most speculative and fragmentary, while later pieces incorporate corrections, confirmations, and contextual background. A chronological feed that surfaces the most recent articles first thus exposes participants to a richer informational environment than one anchored to the initial hours of coverage. This challenges the intuitive assumption that reverse-chronological ordering constitutes a neutral or unbiased baseline; our findings suggest it may represent the most epistemically harmful default in breaking news contexts.

The most robust finding concerns belief calibration rather than accuracy. Epistemic feed participants achieved significantly better calibration than Engagement participants, $t(34.8) = -2.96$, $p = .005$, $d = 0.93$ — a large effect that replicated among readers only ($d = 0.80$, $p = .025$). This finding is consistent with dual-process accounts of media processing [17]: engagement-optimized content, generally designed to be high in affective intensity and conflict framing, activates fast intuitive processing at the expense of the deliberative reasoning needed to accurately assess one's own uncertainty. Crucially, the overconfidence pattern is visible in the NEI selection rate: Engagement participants selected "Not Enough Information" on only 33.7% of questions, compared to 52.1% for Recency participants, and this difference was statistically significant ($p = .044$, $d = 0.68$), yet Engagement accuracy did not differ significantly from Recency. The engagement feed produced participants who felt certain without grounds for certainty—the epistemic profile most resistant to correction. This, in conjunction with Lewandowsky et al.'s [1] continued influence effect, which demonstrates that confidently held misinformation persists even after explicit correction, suggests that engagement-optimized curation may amplify the durability of incorrect beliefs and that these inaccurate beliefs will remain resistant to correction.

The secondary outcome data reveal a finding that is, in some respects, more consequential than the primary results. Participants perceived no significant differences between feed conditions on any of the six perceptual dimensions assessed: trustworthiness, perceived bias, emotional activation, perceived contradictions, charged language, and contextual depth (all $p > .10$). Engagement participants did not rate their feed as more emotionally activating or less trustworthy. In fact, the Engagement condition produced the highest mean trustworthiness rating ($M = 5.00$) of the three conditions, despite generating

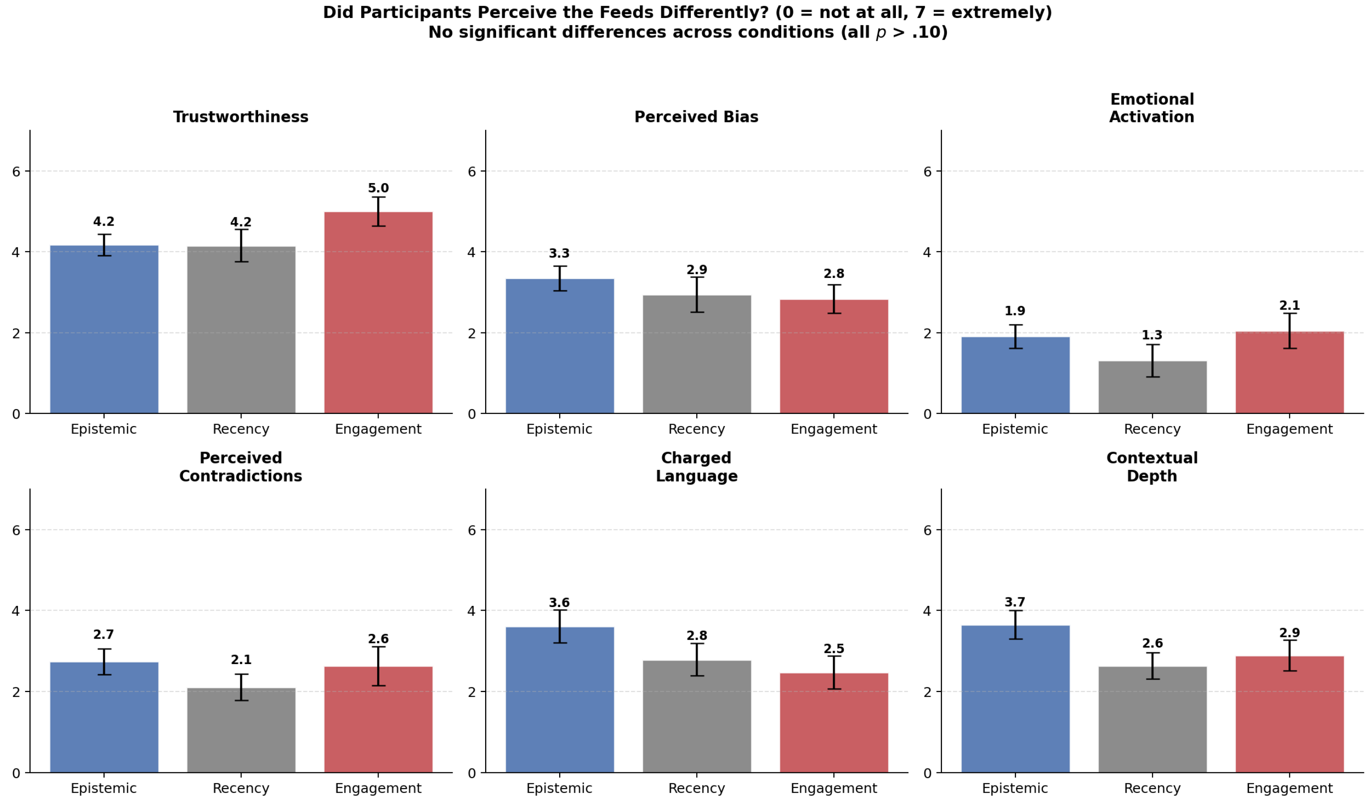

Fig. 3. Mean ratings of perceived feed characteristics by feed condition ($n = 61$). All items rated on a 0–7 scale. Error bars represent ± 1 SEM. No significant differences were observed across conditions (all $p > .10$).

the worst calibration. This dissociation between subjective experience and objective outcome indicates that feed curation effects operate below the threshold of conscious awareness. Users cannot detect that they are being shown differentially curated content, which fundamentally limits the effectiveness of individual media literacy interventions as a corrective mechanism. This finding is consistent with Kramer et al.'s [5] demonstration that the emotional valence of algorithmically curated feeds affects users' own affective states without their awareness, and extends that result to the domain of epistemic outcomes: curation shapes not just how users feel but what they believe they know, again without conscious detection. It also complicates the interpretation of Bakshy et al.'s [3] finding that individual choice plays a larger role than algorithmic curation in limiting exposure diversity: if users cannot perceive that their feed has been epistemically optimized or degraded, they cannot exercise meaningful individual choice to counteract it.

A. *Qualitative Follow-up User Interview*

To complement the quantitative findings, we conducted a brief semi-structured interview with one participant who had completed the survey (age 27, holds a Masters degree, liberal values, gets news primarily from social media). The interview explored how the participant engaged with the feed stimulus and how that engagement compared to their habitual news consumption behavior. While a single interview cannot support generalizable claims, the participant's responses surface several patterns that corroborate and extend the quantitative results.

The participant reported that their engagement with the experimental feed differed substantially from how they normally consume news. In everyday practice, they described looking first at comments to determine whether a story warranted deeper attention. This strategy was unavailable in the experimental feed interface, which may have disrupted their normal information-triage process. They also noted reading articles with a specific goal in mind, seeking answers to particular questions (for example, how the Fico shooting related to broader international political dynamics) rather than absorbing general factual content. In practice, they reported reading only headlines and spending little time on individual articles because the posts appeared repetitive in their factual content.

These self-reported behaviors are consistent with the quantitative finding that overall accuracy was modest across all conditions. The participant themselves expressed surprise at how little they retained about the specific factual details of the event after reading through an entire feed, though they did attribute this partly to their own reading habits. This is consistent with our availability-versus-salience account: even when factual information is technically present in the feed, it may not be absorbed if it does not surface through the specific cues—comments, engagement signals, outlet reputation—that users rely on to direct attention. The participant noted, for instance, that they recognized one outlet (OANN) as right-leaning but found its coverage substantively indistinguishable from AP's reporting on this particular story. The low perceived polarization of the topic from a U.S. audience perspective reduced their motivation to investigate further, suggesting that the absence of salient partisan conflict framing -may have suppressed engagement regardless of feed condition.

Finally, the participant speculated that public responses to the event, such as comments and social media reactions, were likely more emotionally polarizing than the news articles themselves. This observation is particularly relevant to the secondary outcome findings: participants reported low emotional activation across all feed conditions (M range: 1.21–2.15 on a 0–7 scale), and perceived charged language was similarly muted. If the affective content that drives engagement-feed effects operates primarily in the social layer surrounding news articles, rather than in the articles themselves, then experimental paradigms that present articles without their social context may underestimate the magnitude of engagement-curation effects in naturalistic settings. This constitutes both a limitation of the present study and a direction for future work: incorporating social engagement signals into the experimental stimulus may more faithfully replicate the conditions under which algorithmic curation shapes belief formation in practice.

B. Limitations

Several limitations should be noted. First, this is a single-event study; our findings are specific to the Fico shooting and may not generalize to other crisis types. The methodology is designed to be reusable across events, and we identified four candidate events across different domains for future work.

Second, our sample size is constrained by budget limitations. With 61 participants split across three conditions, we are underpowered to detect small effects. We report effect sizes throughout to support interpretation independent of statistical significance. Our data is underpowered for the medium-sized accuracy effects observed, which would require approximately 54 participants per cell to detect reliably at 80% power. Because of the homogeneity described in the "Participants and Exclusions" section of Results, further work remains to expand generalizability to older, less formally educated, or politically conservative news consumers.

Third, the feed environment, while visually realistic, is not a naturalistic media consumption context. Participants knew they were in a study, which may have altered their reading behavior. We mitigated this by not disclosing the existence of multiple feed conditions and by including an attention check.

Fourth, the epistemic feed's weighted random sampling introduces within-condition variation in article exposure, which could increase noise in that condition's outcomes. We record article IDs per participant to enable post-hoc analysis of exposure effects.

Fifth, our article scoring, while validated against human annotations, relies on categorical 1–3 scales that may not capture fine-grained differences. The LLM-assisted scoring, while efficient, introduces a dependency on model behavior that may not be stable across model versions. Furthermore, additional work should be completed to validate model-produced scores, and additional work could be done to improve the codebook to produce better aligned scores for AI and CF.

VI. Conclusion

We presented an experimental framework for measuring how algorithmic feed curation affects factual belief formation following a crisis event. By scoring real news articles on six quality and framing axes and constructing three feeds with formally specified ranking objectives, we demonstrated that engagement-optimized feeds resulted in users being more confidently wrong about factual true/false questions. Additionally, these users rated their feeds as no more trustworthy, biased, or emotionally activating than participants in other conditions. Our pipeline—from GDELT-based corpus construction through codebook-validated scoring to realistic feed deployment with integrated survey instrumentation is fully reproducible and extensible to new events. The complete codebase, scored article data, and deployment files are available at GitHub repository.

These findings have implications for platform design, media literacy, and the normative question of what news feeds *should* optimize for. If engagement-optimized feeds systematically produce overconfident but inaccurate beliefs about crisis events, this represents a measurable epistemic cost that should inform both regulatory and design conversations about algorithmic curation.

Data and Code Availability

The scored article dataset (64 articles with six-axis scores and verified URLs), codebook, feed source code, survey instrument, and Netlify deployment files, survey data, and statistical analysis work (for both this report and the final presentation) are available at in our GitHub repository https://github.com/inthree3/epistemic-feed-study.

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